Alex N. Garza

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Research Interests

Differential privacy, applied cryptography, usable encryption, and usable security. I build privacy protections that keep their guarantees in real implementations and for the people who rely on them.

Education

Tufts University, Medford, MA
Master of Science, Computer Science, expected December 2026
Advisor: Prof. Johes Bater. Member, Tufts Security and Privacy Lab.

The University of Texas at Austin, Austin, TX
Bachelor of Science, 2022

Publications

- **Alex N. Garza**, Johes Bater. *PrivQ: A Python Framework for Modeling Privacy-Preserving Query Processing.* Under submission, 2026. PDF

Software

PrivQ: privacy-preserving SQL query transformation library for Python. Version 1.3.0, 2026. pypi.org/project/privq

Research in Progress

- Furkan Sarıkaya, **Alex N. Garza**, Mark Hempstead, Johes Bater. *DAPE: Database Privacy Evaluation by Automating LLM-Based Attribute Inference Attacks Through Query Interface.* Planned submission, November 2026.
- **Alex N. Garza**, Samuel Sekyere, Daniel Votipka. *Exploring the Relationship Between Developers' Mental Models and Their Interpretation of Static Analysis Results.* Interview study, Fall 2026.

Research Experience

Graduate Researcher, Tufts University, Medford, MA [dates]

- Built PrivQ, a Python framework that makes the cost models of secure multi-party query systems (Shrinkwrap, ORQ) executable, simulating how padded and differentially private intermediate results move through a query plan. Advised by Prof. Johes Bater.
- Co-developing DAPE, which uses an LLM as an adaptive adversary against live query interfaces to measure database privacy through attribute inference attacks. With Prof. Johes Bater, Prof. Mark Hempstead, and Furkan Sarıkaya.

- Designing and running an IRB interview study on how developers interpret GitHub code scanning (CodeQL) alerts. With Prof. Daniel Votipka and Samuel Sekyere.

Teaching

Teaching Assistant, CS 11: Introduction to Computer Science, Tufts University, Fall 2026
- Hold office hours, facilitate labs, and occasionally lecture in Tufts' introductory C++ course. Instructor: Faith A. Ocitti, PhD.

Industry Experience

Chatsi AI, Austin, TX (Remote)
AI Software Engineer, Aug 2025 to Present
- Own the backend and data layer for the product, including database design, system architecture, and the .NET/C# services that turn application data into customer-facing analytics and training data for internal models.
- Led a backend re-architecture from a proprietary framework to FastAPI, with custom tracing and evaluation hooks for observability.

Software Engineer (API and Data), Jan 2025 to Aug 2025
- Designed secure APIs and C# transformation logic that map inconsistent third-party data into a unified internal schema, with fallback handling for missing or malformed fields.

NtelSec, Inc., Austin, TX (Remote)
Software Engineer, Jan 2023 to Jan 2025
- Built AI chatbots with Azure Machine Learning over federally protected documents for a government program, under strict data-handling and compliance requirements.
- Integrated the chatbot with internal access controls to enforce role-based document visibility.

Select Interactive, Dallas, TX
Full-Stack Software Engineering Intern, Jun 2021 to Aug 2021 and Sep 2022 to Dec 2022
- Built responsive React interfaces and integrated them with backend services through REST APIs.

Skills

Research methods: Differential privacy, secure multi-party computation, LLM-driven privacy attacks, qualitative interview studies
Languages: Python, C#, C++, Java, Scala, JavaScript, SQL, HTML/CSS
Tools and platforms: Git, Docker, Azure, GCP, Azure OpenAI, Azure AI Studio
Frameworks: .NET, FastAPI, Node.js, Express, React, LangFuse